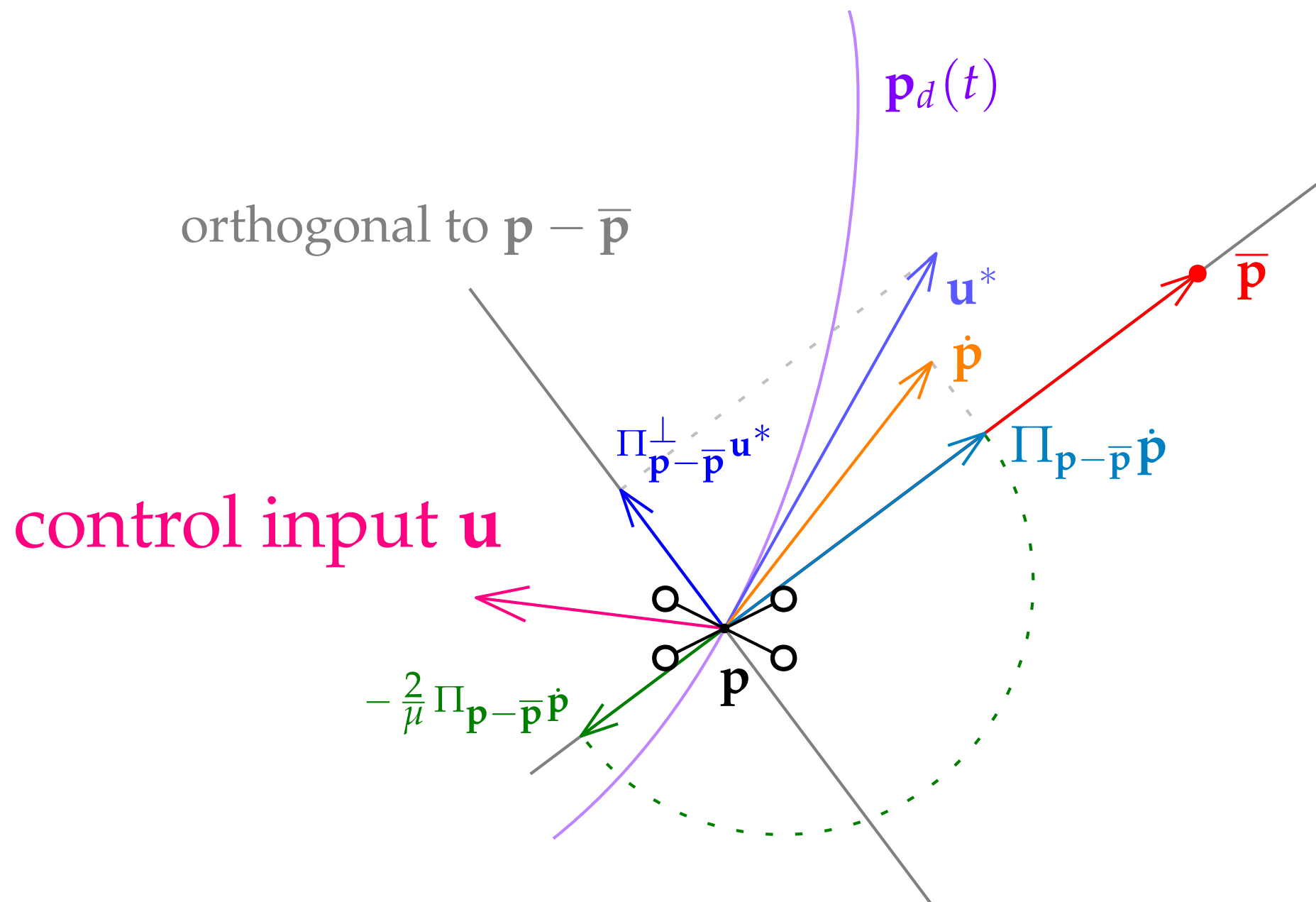
$\mathbf{p}_d(t)$
orthogonal to $\mathbf{p} - \overline{\mathbf{p}}$
$\mathbf{u}^*$
$\dot{\mathbf{p}}$
$\overline{\mathbf{p}}$
$\Pi^{\perp}_{\mathbf{p}-\overline{\mathbf{p}}}\mathbf{u}^*$
$\Pi_{\mathbf{p}-\overline{\mathbf{p}}}\dot{\mathbf{p}}$
control input $\mathbf{u}$
$-\frac{2}{\mu}\Pi_{\mathbf{p}-\overline{\mathbf{p}}}\dot{\mathbf{p}}$
$\mathbf{p}$